E2AM-MemRAG: When Grounding Helps, Hurts, and Costs More—A Controlled Study of Generator-Dependent RAG and Energy-Aware Routing

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Abstract—Retrieval, external memory, and evidence verification are often attached to small language models under the assumption that more grounding will improve reliability. That assumption ignores two coupled questions: whether a generator can use the added evidence under a strict output contract, and whether the resulting quality gain justifies its computational cost. We present E2AM-MemRAG, a controlled study of generator-dependent grounding and energy-aware routing. The study constructs 800 deterministic scenario groups across eight evidence and memory tasks, freezes grouped train/calibration/validation/test splits, and evaluates 17 pilot-retained routes on 120 sealed-test questions. Five pinned FP16 generators are compared directly and under an identical hybrid-retrieval, graph-memory, and deterministic verification route. Success requires answer correctness, complete required-evidence citations, perfect citation precision, correct abstention, parseability, and no forbidden evidence. Generation-window selected-GPU board energy is obtained by integrating 50-ms NVML power samples on NVIDIA T4 GPUs; CPU and whole-system energy are outside this boundary.

Grounding was not uniformly beneficial. It changed strict success by −12.5 percentage points for Qwen3-0.6B, +23.3 points for Granite-4.1-3B, +10.8 points for SmolLM3-3B, and +29.2 points for Qwen3-4B. Every grounded endpoint had a higher observed GPU-energy mean; four pairs were same-board and the Granite-4.0-1B contrast was cross-board. The best route reached 44.2% strict success at 176.68 J/query. The router's validation constraints were infeasible; a disclosed pre-test amendment froze a fail-closed threshold and prohibited a positive routing claim. On test, the policy fell back to the same route for all queries, obtained 0% strict success, and saved 0 J, so the confirmatory hypothesis failed. The result is a mechanism-level negative finding: infeasible calibration constraints and an uninformative fail-closed rule can nullify an otherwise useful route set, and grounding gains must be established per generator rather than assumed.

Index Terms—retrieval-augmented generation, language-model routing, Green AI, GPU energy, long-term memory, abstention, negative results, reproducible evaluation

I. Introduction

Retrieval-augmented generation (RAG) gives a language model access to evidence that is fresher, more specific, or more auditable than its parameters alone [1]. In practice, however, "adding RAG" is not one intervention. A deployed pipeline may decide whether to retrieve, choose sparse or dense search, expand a second hop, consult long-term memory, lengthen the prompt, invoke a larger generator, verify citations, or abstain. Each decision changes both the available evidence and the work performed. A quality-only evaluation can therefore reward a route that is too expensive to deploy, while a cost-only router can learn to choose cheap routes that do not satisfy the evidence contract.

This tension is especially sharp for compact language models. Small generators are attractive for local and resource-constrained inference, but retrieved context is useful only if the generator can identify the relevant facts, follow the requested schema, cite the correct evidence, and withhold unsupported answers. Imperfect or distracting context can actively reduce answer quality [2], [3]. Consequently, the effect of grounding is an interaction among the generator, retrieval mechanism, memory organization, prompt budget, and verifier; it is not a property of the retriever in isolation.

Efficiency introduces a second ambiguity. Prior work has argued that computational cost should be reported alongside predictive quality [4], [5], and inference itself can be a material energy burden [6]. Yet "energy" may refer to a GPU board, a host, an entire cluster, or a carbon estimate. This study uses a deliberately narrow, reproducible quantity: board-power samples from the selected NVIDIA T4 integrated over the call to model.generate(). Retrieval runs on CPU and is charged to latency, not to the reported GPU-energy quantity. We make that boundary explicit because equating GPU board energy with whole-system energy or carbon would be scientifically incorrect.

Model routing appears to offer a natural solution. Cascades and learned routers can trade quality against monetary or computational cost [7]–[9]. For a RAG system, however, a router must choose among composite actions rather than only among model endpoints. Its candidate pool must also contain routes whose success is predictable from information available at decision time. Calibration alone cannot create such separation: a controller that predicts failure for every action can be well calibrated but operationally useless.

We study these issues through E2AM-MemRAG, a frozen controlled experiment rather than a claim about open-domain question answering. HybridBench contains

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800 deterministic scenario groups spanning direct copying, knowledge retrieval, personal memory, knowledge–memory composition, temporal updates, authority conflicts, two-hop retrieval, and deleted or missing evidence. Opaque identifiers prevent parametric memorization of the answers. Grouped splits isolate pilot pruning, router fitting, calibration, threshold selection, and a 120-question sealed test. A strict success indicator requires both answer correctness and exact satisfaction of the applicable citation, abstention, and forbidden-evidence conditions.

The experiment separates three questions. First, a controlled mechanism matrix compares direct generation with BM25, dense, hybrid, and three memory organizations. Second, a five-generator matched panel applies the same direct and grounded route to Qwen3-0.6B, Granite-4.0-1B, Granite-4.1-3B, SmolLM3-3B, and Qwen3-4B-Instruct-2507. Third, a two-stage calibrated router uses only deployable tiny/small routes and is tested against a validation-selected fixed route. All clean test routes cover the same 120 questions, permitting paired, scenario-cluster bootstrap analysis.

The outcome is informative precisely because it is not a routing success. The validation constraints were infeasible: no threshold reached the frozen 80% success floor while satisfying coverage and abstention bounds. Before test labels were accessed, a documented amendment expanded the threshold grid and froze a fail-closed threshold of one, explicitly forbidding a positive hypothesis claim. The resulting policy selected its safe fallback, A03_tiny_hybrid, for all test questions. It achieved zero strict success and reproduced the baseline's 143.16 J/query, so the predeclared energy-reduction hypothesis failed.

The matched generator panel nevertheless exposes a substantial interaction. Grounding changed strict success by −12.5 percentage points for Qwen3-0.6B, +23.3 percentage points for Granite-4.1-3B, +10.8 percentage points for SmolLM3-3B, and +29.2 percentage points for Qwen3-4B. Every grounded endpoint had a higher observed generation-window GPU-energy mean; four of the five pairs were same-board. The small Granite-4.0-1B pair produced no parseable outputs under the frozen prompt/runtime contract, so it is treated as a compatibility failure rather than evidence of general model incapacity.

The paper makes five contributions:

1) a deterministic benchmark that isolates retrieval, memory, temporal, authority, abstention, and citation requirements under grouped leakage controls;
2) a matched five-generator design that distinguishes generator-dependent grounding effects from a flat model leaderboard;
3) an explicit generation-window GPU-energy protocol with complete telemetry, paired latency measurement, and no carbon or whole-system extrapolation;
4) a calibrated composite-action routing protocol with test sealing, abstention and coverage constraints, immutable artifacts, and fresh-root restore;
5) a reproducible negative result identifying validation infeasibility, safe-route collapse, context-sensitive tiny-model behavior, a model-format incompatibility, and robustness floor effects.

The central conclusion is bounded but consequential: having a useful action is not enough if the frozen calibration constraints and decision rule cannot certify or select it. Here, sealed-test analysis later showed that direct route A00 was both more accurate and lower-energy than the universal fallback A03, even though no validation threshold satisfied the predeclared 80% success floor together with coverage and abstention constraints. Before an energy-aware RAG controller is optimized, each generator's ability to use the grounding contract and the feasibility of the selection rule must be established; operational infeasibility must remain a reportable outcome rather than an exception to suppress.

II. Related Work

A. Retrieval-Augmented and Adaptive Generation

RAG combines a parametric generator with retrieved non-parametric evidence [1]. Sparse retrieval remains a strong and interpretable baseline; our lexical routes implement the BM25 probabilistic relevance formulation [10]. Dense passage retrieval and sentence-level bi-encoders provide complementary semantic signals [11], [12]. The hybrid routes in this study fuse sparse and dense ranks and then perform a bounded deterministic second hop, rather than adding an independently trained reranker.

Always-on retrieval is not universally optimal. FLARE initiates retrieval when a model encounters uncertain future content [13]; Adaptive-RAG routes questions among no-retrieval, single-step, and iterative strategies based on estimated complexity [14]; and Self-RAG learns retrieval and critique tokens to regulate evidence use [15]. These systems motivate conditional grounding, but they do not establish that a compact generator receives a positive quality–energy return from a particular evidence contract. E2AM-MemRAG instead freezes the generator-route combinations and measures their paired effects before evaluating a router.

B. External Memory and Evidence Quality

External memory extends the evidence problem from documents to persistent, time-indexed state. LongMem separates a frozen backbone from a long-term memory network [16]; MemoryBank studies acquisition, retrieval, updating, and forgetting [17]; and MemGPT frames tiered memory management as an operating-system problem [18]. Our flat, hierarchical, and graph/temporal routes are not new memory architectures. They are controlled mechanisms for testing whether organization, recency, entity links, and tombstones change strict grounded success under a fixed context budget.

Evaluation must distinguish answer plausibility from evidential support. KILT couples knowledge-intensive tasks with provenance [19], while ALCE evaluates citation correctness and completeness for generated answers [20]. In the present benchmark, citations are exact identifiers in a generated evidence universe. The guard checks parseability, retrieved-set membership, exact required-evidence recall, precision, and forbidden evidence; it is deterministic and is not described as semantic entailment.

Imperfect retrieval can introduce stale, conflicting, irrelevant, or adversarial context [3], and additional context may be detrimental even when it appears topically related [2]. These observations motivate our four corruption conditions and the decision to report abstention and clean-baseline efficacy alongside robustness deltas.

C. Routing, Calibration, and Abstention

FrugalGPT showed that cascades can reduce API cost while preserving or improving task quality [7]. Hybrid LLM learns a quality-aware router between local and remote models [8], and RouteLLM uses preference data to route between stronger and weaker endpoints [9]. Recent benchmark evidence further shows that router rankings depend strongly on the model pool and that sophisticated controllers can approach simple baselines when the pool is poorly separated [21]. These works primarily optimize monetary or model selection cost. Our action is a composite of generator, retrieval, memory, and verification choices, and the measured cost is physical GPU-board energy during generation.

The controller combines probability calibration with a reject option. Post-hoc calibration is essential when probabilities drive operational decisions [22], while selective prediction formalizes the trade-off between coverage and risk [23]. We therefore report calibration, execution coverage, and abstention separately. A low Brier score at near-zero confidence is not interpreted as useful discrimination when all selected test outcomes fail.

D. Energy Reporting for Machine Learning

Green AI argues that efficiency should be a first-class evaluation dimension [4]; related work documents the environmental and policy importance of computationally intensive NLP [24] and calls for systematic energy and carbon reporting boundaries [5]. Deployment-time inference is itself a substantial and workload-dependent energy source [6]. Zeus demonstrates energy–time Pareto reasoning for GPU workloads [25], although its primary setting is training.

Our measurement uses NVIDIA Management Library board-power queries [26] and trapezoidal integration. The quantity does not include the host, CPU retrieval, storage, network, or cooling, and it is not converted to carbon. This narrower claim improves reproducibility but also limits what the energy results can establish.

III. HybridBench: Controlled Evidence and Memory Tasks

A. Design Objective

HybridBench is designed for mechanism identification, not for estimating open-domain QA performance. Its answers are opaque facts sampled independently of task class and split. A model therefore cannot solve an evidence-required item from world knowledge, and a task-only majority rule cannot exploit a fixed answer. The generator is deterministic and does not call an LLM during dataset construction.

Each of the 800 scenario groups creates a project identifier, a direct-copy token, a dataset alias, a private label, an optimizer, scheduler, seed, batch-size update, and an authority conflict. It emits seven knowledge documents: a current optimizer specification, old and new batch-size revisions, approved and unofficial learning-rate records, a project-to-dataset alias, and a dataset-to-scheduler record. It also emits three memory events: a chosen seed, a private label, and a tombstone deleting that label. Tombstoned events are removed from active retrieval. Every query is associated with the exact required and forbidden evidence IDs.

B. Eight Task Classes

The benchmark contains 100 scenarios in each class:

1) no retrieval: reproduce an opaque token supplied in the question;
2) knowledge only: recover the optimizer from an official document;
3) memory only: recover a previously selected seed;
4) knowledge plus memory: compose optimizer and seed;
5) temporal update: prefer the newest approved batch size;
6) authority conflict: prefer a signed high-authority record over a newer unofficial note;
7) two-hop hybrid: traverse project $\rightarrow$ dataset alias $\rightarrow$ scheduler;
8) deleted or missing: abstain because the relevant private fact was tombstoned and is not valid evidence.

The classes separate evidence availability from evidence resolution. For example, temporal and authority tasks require a model to select among plausible conflicting records, whereas the deletion task requires it to avoid both retrieval and citation of forbidden state.

C. Grouped Splits and Leakage Controls

Scenario groups are partitioned 10%/50%/15%/10%/15% into pilot, train, calibration, validation, and sealed test, respectively. This yields 80 pilot, 400 train, 120 calibration, 80 validation, and 120 test questions, with 15 test questions per task class. A scenario ID appears in exactly one split. Each task–split pair uses a distinct template family, and temporal/entity histories remain inside the group.

The leakage audit rejects cross-split scenario overlap, template-family overlap, exact normalized duplicates, and question-shingle Jaccard similarity at or above 0.95. Facts are randomly sampled instead of cycled by task or split. As a construction sanity check, a task-only predictor formed from train-set answer majorities answered 21 of the 90 evidence-requiring test items (23.33%), below the frozen 45% rejection threshold. Pilot outcomes alone may remove infeasible non-anchor routes. Router models use train outcomes; isotonic calibration uses the calibration split; threshold and safe-route selection use validation. Test answer fields and evidence labels remain procedurally sealed until the policy, thresholds, and analysis plan are frozen. Because collaborators share the dataset credential, this is a procedural seal and not a cryptographic access-control claim.

D. Strict Outcome Contract

Model output must parse as a JSON-like object with an answer, a list of exact evidence IDs, and an abstention flag. Let F_i be token F1 between the predicted and expected answer, R_i required-evidence citation recall, P_i citation precision, A_i agreement with the required abstention decision, and V_i an indicator for a parseable output with no citation outside the retrieved set and no forbidden evidence. Strict success is

$$S_i = \mathbb{1}\big[F_i \geq 0.8 \wedge R_i = 1 \wedge P_i = 1 \wedge A_i = 1 \wedge V_i = 1\big]. \tag{1}$$

For tasks without required evidence, recall is one and precision is one only when the citation set is empty. For the deleted/missing class, a correct abstention or an explicit insufficient-evidence answer receives answer credit, but exposure of a tombstoned ID fails the guard. The auxiliary score $0.75F_i + 0.25R_i$ is retained for diagnosis; it does not replace Eq. (1) in any primary analysis.

This contract is intentionally stricter than exact-answer accuracy. It distinguishes three failure layers: output parsing, answer content, and evidence support. It also prevents a fluent unsupported answer from counting as success.

IV. E2AM-MemRAG Methodology

A. Frozen Model Panel

Table I lists the five generators. Exact repository commits are frozen to prevent upstream changes during parallel execution. Qwen3-0.6B and Granite-4.0-1B form the deployable online pair; they must coexist on a single T4 without CPU/disk offload and leave at least 15% free VRAM. Granite-4.1-3B, SmolLM3-3B, and Qwen3-4B are sequential reference models and are never router actions. References are loaded alone and unloaded before the online pair is restored. The model families are documented by their technical reports or exact model cards [27]–[32].

All models use FP16, deterministic greedy decoding (do_sample=False), a maximum of 80 new tokens, and a 1,536-token context budget. Thinking is disabled for checkpoints that expose that template option. A context overflow or OOM is recorded as failure; the runtime does not silently alter precision, batch size, context, or device placement.

B. Retrieval and Memory Mechanisms

The sparse index implements BM25 with $k_1 = 1.5$ and $b = 0.75$. Dense retrieval uses the pinned sentence-transformers/all-MiniLM-L6-v2 checkpoint with explicit attention-mask mean pooling and ℓ_2 normalization on CPU. Dot product ranks the normalized embeddings. Hybrid knowledge retrieval applies reciprocal-rank fusion (RRF),

$$\mathrm{RRF}(d) = \sum_{m \in \{\mathrm{BM25}, \mathrm{dense}\}} \frac{1}{60 + r_m(d)}, \tag{2}$$

then expands only dataset aliases observed in the strongest first-hop evidence. The final evidence budget remains $k = 4$; the second hop cannot widen the context.

Memory uses the same sparse and dense signals. A flat route returns dense neighbors. The hierarchical route fuses rankings and retains the newest active event per entity. The graph/temporal route expands entity-linked history, removes tombstoned events, and ranks valid timestamps ahead of missing or malformed ones, followed by relevance and a deterministic ID tie-break. These mechanisms test organization and recency; they are not claimed as new learned memory architectures.

C. Route Matrix and Evidence Guard

Twenty-two routes were declared. A00–A15 span direct, sparse, dense, hybrid, flat-memory, hierarchical-memory, graph-memory, combined, and evidence-guarded mechanisms. M16–M21 complete the five-model matched panel. Pilot-only pruning removed A05–A07, A11, and A15; all ten direct/grounded endpoints remained mandatory. The sealed clean matrix therefore contains 17 routes × 120 questions.

The matched grounded route is logically identical across generators: bounded hybrid knowledge, graph/temporal memory, the same evidence budget, and a deterministic guard. After generation, the guard rejects citations outside the retrieved set and rejects a retrieved-context answer that supplies no citation. A rejection is converted to an insufficient-evidence abstention before strict scoring. This is a syntactic and set-membership guard, not an LLM entailment judge.

D. Generation-Window GPU-Energy Measurement

Immediately before model.generate(), CUDA is synchronized and an NVML sampler starts at a 0.05-s interval. After generation, CUDA is synchronized again and sampling stops. One route-specific warm-up execution absorbs model loading and tokenizer initialization and is discarded before canonical rows are measured. No idle-power subtraction is applied. Given board-power samples (t_j, P_j), energy is estimated by trapezoidal integration,

$$E_i = \sum_{j=1}^{n_i - 1} \frac{P_j + P_{j+1}}{2} (t_{j+1} - t_j). \tag{3}$$

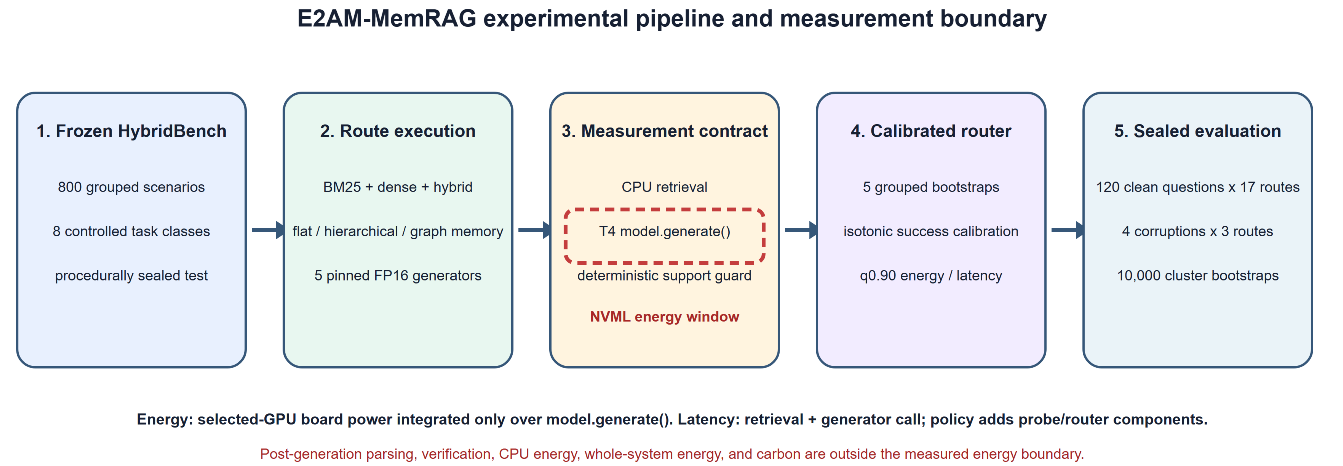

Fig. 1. Experimental pipeline and measurement boundary. The red dashed region is the only interval included in reported GPU energy. CPU retrieval and post-generation checking are not assigned an energy value.

TABLE I
Frozen generator panel and matched direct/grounded endpoints. Only the first two model families were eligible for online routing. All checkpoints were loaded in FP16 at immutable revisions.

Role	Frozen repository (revision prefix)	Direct	Grounded/verified	Router eligible
Tiny	Qwen/Qwen3-0.6B (c1899de)	A00	M16	yes
Small	ibm-granite/granite-4.0-1b (6a7381b)	A04	A13	yes
Granite reference	ibm-granite/granite-4.1-3b (c065040)	M17	M18	no
Peer reference	HuggingFaceTB/SmolLM3-3B (a07cc9a)	M19	M20	no
Upper reference	Qwen/Qwen3-4B-Instruct-2507 (cdbee75)	M21	A14	no

At least two samples and a valid GPU UUID are required. Missing or non-finite energy blocks release-level inference rather than disappearing from a mean. Hub uploads occur outside the measured block.

Equation (3) measures the selected board during generation only. Dense encoding and retrieval run on CPU; prompt construction, parsing, evidence verification, and scoring are outside the energy window. Route latency is retrieval time plus the duration of the generator call. Policy latency adds the separately timed query/probe router components. Neither quantity is labeled whole-system energy, full end-to-end energy, or carbon.

E. Two-Stage Calibrated Router

The router considers only pilot-retained online routes. Stage 0 uses eight query-available features, including length, token/number counts, temporal/conflict terms, and memory cues, to score direct actions. If none clears the constraints, one charged BM25/memory-metadata probe supplies eight retrieval features such as top scores, score gaps, counts, authority, and recency. Stage 1 scores non-direct actions and otherwise chooses the validation-selected safe route.

For each of five grouped bootstrap seeds, a histogram gradient-boosting classifier predicts success. Isotonic regression calibrates it on the calibration split. Separate histogram gradient-boosting regressors predict log energy and log latency with quantile loss at $q = 0.90$. For action a, the conservative success estimate is the minimum calibrated probability across seeds,

$$\underline{p}_a(x) = \min_{b\in\{1,\ldots,5\}} \widehat{p}^{\,\mathrm{cal}}_{a,b}(x), \tag{4}$$

while cost estimates are seed means of exponentiated quantile predictions. Among actions satisfying $\underline{p}_a(x) \geq \tau$ and predicted latency at most 12 s, the controller chooses the lowest predicted generation-window GPU energy. Reference models are excluded even when they lie on the offline Pareto frontier.

V. Experimental Protocol

A. Execution and Provenance

The study ran in Kaggle GPU sessions with exactly one visible NVIDIA T4 per scientific worker. A dual-T4 allocation was permitted, but the device mask was set before importing CUDA-aware packages. Stage 00 froze source/configuration hashes, model and encoder revisions, prompts, route specifications, decoding parameters, seed 4622, and the Python/Torch/CUDA environment. Each later stage rejected a changed experiment ID, specification hash, source bundle, environment contract, or worker lane.

Large model snapshots were verified inputs and were not uploaded as results. Replayable traces and checkpoints were written as immutable content-addressed objects. A worker pointer was published only with a closure whose dependencies already existed remotely. The final release was restored into a fresh root and verified by SHA-256 before the completion marker was published. Completion was kept separate from the hypothesis result.

The pilot evaluated all 22 declared routes on 80 questions. The 17-route clean matrix retained all mandatory anchors and pruned only five nonanchors using the pilot rule. Router training then used 11 deployable tiny/small routes across 400 train, 120 calibration, and 80 validation queries: 4,400, 1,320, and 880 route-query traces, respectively. The test matrix was not used for fitting, calibration, thresholding, or safe-route selection.

B. Validation Infeasibility and Protocol Amendment

The original frozen threshold grid was $\{0.60, 0.65, 0.70, 0.75, 0.80, 0.85\}$. A threshold was feasible only if validation strict success was at least 0.80, execution coverage at least 0.90, and abstention at most 0.20. Every original candidate produced 8.75% success, 100% execution coverage, and 18.75% abstention. Thus no threshold was feasible, and the original implementation stopped before creating a policy artifact.

After observing validation infeasibility but before accessing test labels, the study recorded amendment stage06-threshold-completion-separation-v1. It extended the audit grid to 0.90, 0.95, and 1.00; all remained infeasible with the same validation outcomes. The amendment then froze $\tau = 1.0$ as a fail-closed completion policy, recorded validation_feasible=false, and explicitly set hypothesis_claim_allowed_when_infeasible=false. This preserved the distinction between completing the frozen test matrix and claiming router success. We report the amendment because describing the eventual fallback as an ordinary validation-selected operating point would be misleading.

The safe route itself was selected on validation by highest strict success, then lowest median measured GPU energy, and then route ID. Under the fail-closed policy, actions unable to clear the conservative threshold invoke that safe route.

C. Confirmatory Estimands and Gates

For test question q, let $\pi(q)$ be the frozen policy route and b the fixed safe baseline. The paired success and energy estimands are

$$\Delta_S = \frac{1}{N} \sum_{q=1}^{N} \left(S_{q,\pi(q)} - S_{q,b} \right), \tag{5}$$

$$\Delta_E = \frac{1}{N} \sum_{q=1}^{N} \left(E_{q,\pi(q)} - E_{q,b} \right). \tag{6}$$

The confirmatory claim required all four conditions:

$$\begin{aligned} L_{.95}(\Delta_S) &\geq -0.03, \quad & U_{.95}(\Delta_E) &< 0, \\ \text{coverage} &\geq 0.90, \quad & \text{abstention} &\leq 0.20. \end{aligned} \tag{7}$$

Quality non-inferiority alone is not sufficient. In particular, equality at zero success can pass the relative margin without preserving useful quality.

D. Uncertainty and Secondary Analyses

Intervals use 10,000 percentile bootstrap replicates with seed 4622. Each replicate resamples whole scenario IDs and retains paired route outcomes. Because every test scenario contributes one query, this is a paired nonparametric bootstrap over 120 scenarios. The 2.5th and 97.5th percentiles form the reported interval. No p-values are computed.

Secondary analyses estimate matched grounded-minus-direct effects for success, energy, latency, citation recall, and abstention; route-level mechanism contrasts; and condition-minus-clean success effects. These intervals are exploratory and are not multiplicity-adjusted. They quantify variation across the frozen questions, not stochastic decoding or repeated hardware runs.

E. Clean and Robustness Evaluation

Every retained clean route executes all 120 test questions once with deterministic decoding, producing 2,040 trace rows. Route execution is route-major rather than randomized or interleaved. The robustness experiment evaluates three deployable anchors—A00 direct, A03 hybrid, and A13 grounded/guarded—under stale evidence, untrusted conflict injection, missing evidence, and deletion/duplicate ordering. This produces $3 \times 4 \times 120 = 1{,}440$ rows.

The corruption conditions were executed on separate T4 boards. Their energy means are therefore not compared across conditions. A substring-based compromise flag detects the literal marker used by the synthetic injection; it is not a general prompt-injection evaluator.

F. Hardware Comparability

Clean routes were distributed across four physical T4 boards. Four matched model pairs—Qwen3-0.6B, Granite-4.1-3B, SmolLM3-3B, and Qwen3-4B—used the same board within each pair. Granite-4.0-1B direct and grounded endpoints used different boards. Tiny BM25/dense/hybrid contrasts were same-board; the tiny memory routes were on a different board from A00. Accordingly, within-board paired energy effects receive the clearest interpretation. Cross-board, cross-model, and memory-versus-direct energy differences are descriptive and may include session, thermal, residency, and board effects. The primary policy–baseline comparison is unaffected because both sides reference the exact same A03 trace row.

VI. Results

A. Release Integrity

The experiment completed and the release passed a fresh-root restore. All 11 Stage-09 artifacts matched their declared SHA-256 hashes and byte counts. The clean set contains 2,040 unique units and the robustness set 1,440; both have zero duplicate IDs, divergent route hashes, non-finite audited values, or execution failures. Generation-window energy coverage is 100% in both sets (Table II). Thus the negative outcome is not attributable to missing runs or telemetry filtering.

Across all clean routes, 58.38% of outputs parsed, 19.41% passed the answer check, 18.04% passed the support guard, and 11.23% met every strict condition. These aggregate rates are diagnostic rather than model-weighted benchmark scores because the route matrix deliberately repeats every question under many mechanisms.

B. Confirmatory Router Outcome

The frozen policy invoked A03 for all 120 questions. Its strict success was 0% (95% CI [0, 0]), mean generation-window GPU energy was 143.16 J/query ([138.14, 148.07]), and the summed policy-latency components averaged 3.760 s/query ([3.652, 3.863]). Execution coverage was 100% and abstention 15%.

The fixed baseline was also A03. Consequently, both paired differences were exactly zero: $\Delta_S = 0$ and $\Delta_E = 0$ with degenerate intervals. The formal relative quality and operating gates passed, but the upper energy bound was not below zero. The joint hypothesis therefore failed. The quality result is floor equivalence, not preservation of useful quality.

Test confidence occupied a single 0–0.1 bin, with mean 0.00937, Brier score 8.79×10^{-5}, and 10-bin ECE 0.00937. Because every selected outcome was zero, these small errors indicate a correctly pessimistic but non-discriminating policy; they are not evidence of broadly effective calibration.

C. Route-Level Quality–Energy Structure

Figure 2 shows that the best route was the offline Granite-4.1-3B grounded endpoint M18: 44.17% success at 176.68 J/query and 2.647-s median route latency. The grounded Qwen3-4B reference A14 achieved 42.50% at 256.13 J, while grounded SmolLM3 M20 achieved 34.17% at 148.93 J. Among direct references, M19 reached 23.33%, M17 20.83%, and M21 13.33%. The only nonzero online tiny/small direct endpoint was A00 at 12.50% and 54.71 J.

The grounded Pareto designation in the release applies only to the five grounded model endpoints in success–energy–latency space; it identifies the Granite-4.1-3B and SmolLM3 families. It is not a global frontier over all 17 routes, and the best grounded points were not deployable policy actions.

D. Generator-Dependent Grounding

The matched panel in Fig. 3 and Table V rejects a model-independent account of grounding. Within the equal eight-task mixture, Qwen3-0.6B changed from 12.50% direct success to zero grounded success (-12.50 percentage points, interval $[-18.33, -6.67]$). That aggregate loss came entirely from the no-retrieval stratum: A00 copied all 15 supplied tokens, whereas M16 copied none. Both routes already scored zero on every evidence-required stratum. It is therefore inaccurate to claim that RAG reduced this model's success on questions that needed evidence; the experiment instead shows that an always-grounded policy destroyed its one functioning direct behavior without creating evidence-task success.

Granite-4.1-3B improved from 20.83% to 44.17% (+23.33 percentage points, [10.00, 37.50]), and Qwen3-4B improved from 13.33% to 42.50% (+29.17 percentage points, [16.67, 41.67]). SmolLM3 increased from 23.33% to 34.17%, but its interval $[-2.50, 24.17]$ includes zero. Positive aggregate effects arose mainly from knowledge, memory, temporal-update, and authority-conflict tasks. The same grounded route frequently lost direct-route successes on no-retrieval and deleted/missing tasks. Hence the estimand is "always grounded minus always direct" under the frozen equal task mixture, not the benefit of retrieving only when retrieval is needed.

All five grounded endpoints had higher observed generation-window GPU energy. The same-board increases were 146.72 J for Qwen3-0.6B, 119.53 J for Granite-4.1-3B, 41.52 J for SmolLM3, and 171.71 J for Qwen3-4B. Grounding also increased mean input tokens substantially: for example, Granite-4.1-3B rose from 128.1 to 627.8 tokens, and Qwen3-4B from 128.4 to 760.0. These are per-assigned-query effects under the frozen prompts, not intrinsic or token-normalized model-efficiency rankings.

E. The Granite-4.0-1B Compatibility Failure

Granite-4.0-1B scored zero both directly and grounded. Inspection of all clean A04, A12, and A13 rows showed an identical 80-character sequence of exclamation marks; every output reached the 80-token ceiling and none parsed. The A04 and A13 energy endpoints also ran on different T4 boards. We therefore treat this pair as an observed frozen prompt/chat-template/runtime incompatibility. It removed one candidate from the useful online pool, but neither explains the policy-level calibration infeasibility by itself nor shows that the model family is generally incapable of RAG.

F. Retrieval, Memory, and Failure Layers

On the same board, A00 achieved 12.5% while tiny BM25, dense, and hybrid routes each achieved zero. Their measured energy increases were 102.97, 95.70, and 88.44 J/query, respectively, with intervals excluding zero. Again, the success difference is the loss of no-retrieval cases rather than a decline from positive evidence-task performance.

TABLE II
Released trace integrity and outcome audit. Both sets had zero duplicate unit IDs, zero non-finite audited values, zero divergent route hashes, and zero execution failures.

Set	Rows	Routes	Strict successes	Parse	Support	Energy coverage
clean	2040	17	229 (11.2%)	58.4%	18.0%	100.0%
robustness	1440	3	60 (4.2%)	49.8%	6.7%	100.0%

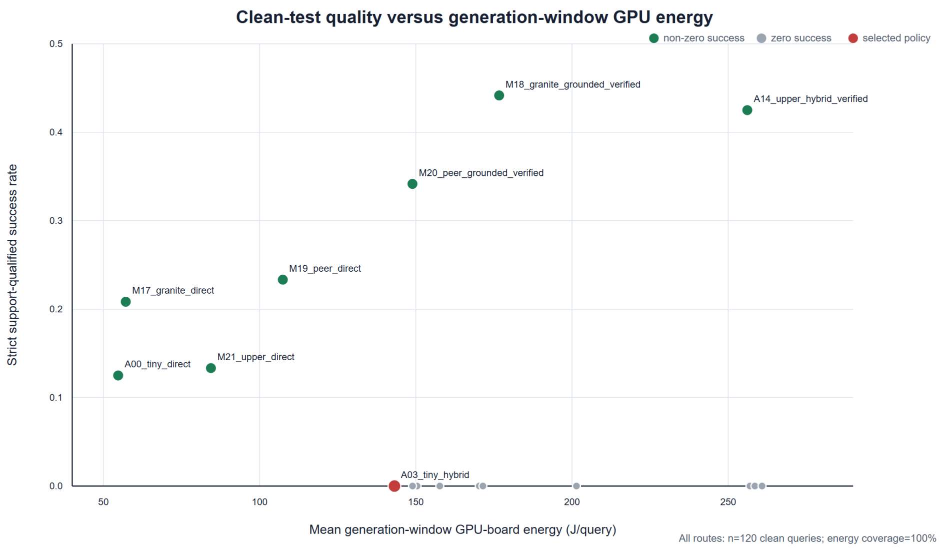

Fig. 2. Strict success versus generation-window selected-GPU board energy for the 17 retained clean routes. Offline reference routes provide the best quality points but were not eligible router actions. Cross-board comparisons are descriptive; Table VIII gives route assignment.

TABLE III
Confirmatory policy result on the sealed test split ($n = 120$). Intervals are 95% scenario-cluster bootstrap intervals.

Quantity	Result
Selected route	A03 for all queries
Strict success	0.0% [0.0, 0.0]
Generation-window GPU energy	143.16 J/query
Energy interval	[138.14, 148.07] J
Policy latency component sum	3.76 s/query
Abstention / execution coverage	15.0% / 100.0%
Policy − baseline success	0.0 pp
Policy − baseline energy	0.00 J/query
Quality non-inferiority	pass
Energy reduction	fail
Operating constraints	pass
Joint confirmatory hypothesis	fail

TABLE IV
Selected clean-test route results. GPU J denotes mean generation-window selected-board energy; the complete matrix and hardware assignment are in Appendix A.

ID	Success (%)	GPU J	Median s
M18	44.2	176.68	2.65
A14	42.5	256.13	3.79
M20	34.2	148.93	2.34
M19	23.3	107.42	1.67
M17	20.8	57.14	0.45
A00	12.5	54.71	1.44
A03	0.0	143.16	3.47

Tiny flat, hierarchical, and graph-memory routes also scored zero and showed larger energy means than A00, but they ran on a different board; those energy differences are descriptive. Across the common grounded route, retrieval recovered the complete required evidence for most evidence-requiring queries, yet success remained generator-dependent. For example, two-hop evidence was frequently retrieved while Granite-4.1-3B and SmolLM3 still obtained zero strict success on that class. Retrieval completeness was therefore necessary but not sufficient.

Failure decomposition reinforces this point. M18 parsed

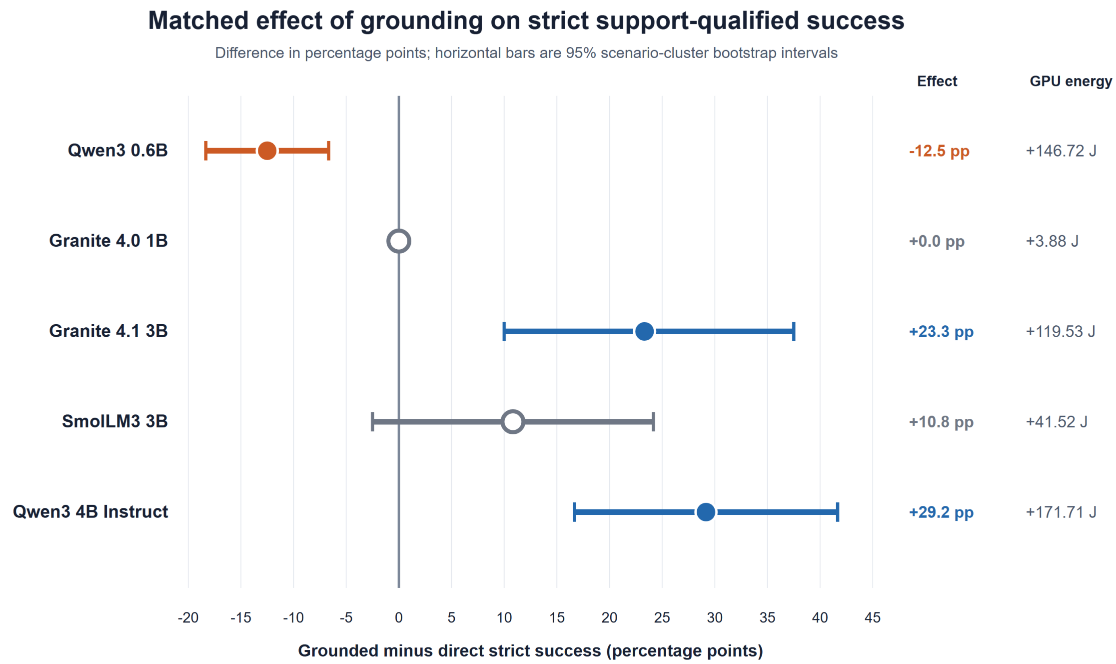

Fig. 3. Matched grounded-minus-direct strict-success effects. Filled markers denote intervals that exclude zero; the open SmolLM3 marker spans zero. The energy column reports the paired generation-window GPU-energy increase. The Granite-4.0-1B energy pair is cross-board and descriptive.

TABLE V
Matched direct-to-grounded transfer on the same 120 sealed-test questions. Success effects are percentage points; energy effects are generation-window selected-GPU board joules per query. "Same board" indicates whether both endpoints ran on the same physical T4.

Generator	Direct (%)	Grounded (%)	Δ success [95% CI]	Δ GPU J [95% CI]	Same board
Qwen3 0.6B	12.5	0.0	-12.5 [-18.3, -6.7]	+146.72 [+141.56, +151.62]	yes
Granite 4.0 1B	0.0	0.0	+0.0 [+0.0, +0.0]	+3.88 [+1.73, +6.14]	no
Granite 4.1 3B	20.8	44.2	+23.3 [+10.0, +37.5]	+119.53 [+108.73, +129.74]	yes
SmolLM3 3B	23.3	34.2	+10.8 [-2.5, +24.2]	+41.52 [+35.47, +47.50]	yes
Qwen3 4B Instruct	13.3	42.5	+29.2 [+16.7, +41.7]	+171.71 [+157.74, +185.24]	yes

96.67% of outputs, passed the support guard on 74.17%, and succeeded on 44.17%. M20 parsed 100%, passed support on 55.83%, and succeeded on 34.17%. A14 parsed 84.17%, passed support on 67.50%, and succeeded on 42.50%. In contrast, A03 parsed 46.67% but had zero support-qualified outputs. Sixty-four A03 outputs failed parsing and 40 contained citation representations that did not satisfy exact ID membership. The frozen interface contract, not retrieval alone, was a major bottleneck.

G. Robustness Is a Floor Effect

The selected policy used A03 for every query under all four corruptions and achieved zero strict success, matching its clean zero. The resulting zero degradation is uninformative: it reflects a floor, not retained capability. A00 remained at 12.5% because it ignores evidence, while A13 remained at zero because of the compatibility failure.

TABLE VI
Selected-policy outcomes under four evidence corruptions. Zero clean and corrupted success makes every success delta a floor effect.

Condition	Success (%)	Abstain (%)	Coverage (%)
conflict injection	0.0	15.8	100.0
deletion duplicates	0.0	15.0	100.0
missing	0.0	18.3	100.0
stale	0.0	15.0	100.0

The synthetic compromise marker was never emitted, but a zero marker rate alongside zero policy efficacy cannot support a broad prompt-injection robustness claim.

VII. Failure Analysis and Design Implications

A. Route Viability Precedes Router Optimization

The router failure was not simply a poor choice of τ. Validation success was 8.75% at every candidate threshold from 0.60 through 1.00. One deployable family had no evidence-task success; the other produced a repeated, unparsable sequence under the frozen integration. No calibration method can create a useful operating point when every feasible action is weak or operationally incompatible. A practical router-development pipeline should therefore impose a precondition: at least two deployable actions must demonstrate nontrivial, query-dependent strict success before policy fitting begins. If the condition fails, the correct output is an infeasible action pool, not a forced router leaderboard entry.

The exact-matrix evaluation also clarifies what happened on test. The policy did not prospectively discover that A03 was optimal for every question; it invoked the same safe fallback because no predicted action cleared the fail-closed threshold. This distinction matters when diagnosing routing systems and when comparing them with simple baselines [21].

B. Grounding Is a Generator–Task Interaction

The aggregate transfer effects conceal sharp task interactions. Always grounding helped the 3B/4B generators on several evidence-requiring classes while removing some direct successes on no-retrieval and deleted/missing cases. For the 0.6B model, it removed the only successful stratum and failed to unlock any other one. An adaptive system therefore needs both a capable grounded route and a reliable retrieval-need signal. "Small model" is too coarse an explanatory variable; prompt compliance, context use, instruction tuning, tokenizer/chat templates, and task structure jointly determine whether evidence becomes useful.

This finding also limits the common practice of reporting one RAG-minus-direct average. Such an average depends on the task mixture. Under traffic dominated by knowledge or temporal questions, the larger grounded endpoints would look more favorable; under direct-copy or valid-abstention traffic, always-grounded routes would look worse. Deployment claims require either a representative workload or stratified reporting.

C. The Interface Is Part of the System

Strict success exposed output-interface failures that answer-only evaluation would blur. The Granite-4.0-1B repetition failure occurred before semantic evaluation. A03 often produced superficially citation-like values that did not match exact retrieved IDs. The deterministic guard then rejected unsupported or malformed outputs. These are not cosmetic parser errors: a production evidence system needs machine-actionable citations and predictable abstention. Still, exact-ID scoring can be harsher than human recoverability, so future work should separate schema conformance from semantic citation entailment instead of treating either as the other.

The observed failure suggests a mandatory compatibility gate per model–prompt–runtime triple: validate the chat template, schema adherence, termination behavior, and citation serialization before collecting expensive route matrices. A compatibility failure should create a new versioned experimental specification rather than a silent prompt repair after test access.

D. Measured Energy Is Not Total RAG Energy

Every grounded endpoint increased the energy observed during generation, consistent with longer prompts and, frequently, longer outputs. This does not imply that the reported difference equals the total energy cost of RAG. CPU dense encoding, lexical search, memory traversal, routing, and verification are absent from the energy quantity. Nor are host idle power, storage, cooling, and network transfers. The result supports a narrower statement: under these frozen routes and same-board matched pairs, generation itself consumed more selected-board energy when the grounded prompt was used.

Future evaluations should add synchronized CPU/host power measurement, idle-power subtraction where appropriate, repeated interleaved trials, and an explicit boundary diagram. Reporting per-query energy together with token counts, latency, and useful successes would make overhead and output-length effects more legible.

E. Robustness Requires a Nonzero Clean Reference

A zero corruption delta is meaningful only when the clean system performs the task. Here the selected route began at zero, so the corruption study cannot distinguish stable failure from robust competence. A better robustness gate would first require a minimum clean utility for each evaluated anchor, then report both absolute corrupted success and paired degradation. Attack-specific compromise metrics should likewise be presented as narrow detectors, not substitutes for task success or general adversarial robustness.

F. Actionable Recommendations

The evidence supports five concrete design rules:

1) gate model/chat-template/schema compatibility before route benchmarking;
2) establish generator- and task-specific grounding value before training a controller;
3) treat validation infeasibility as a first-class result with a fail-closed deployment state;
4) measure and label energy boundaries precisely, and avoid causal energy comparisons across unbalanced boards or sessions;
5) require nonzero clean utility before interpreting robustness deltas.

VIII. Limitations, Reproducibility, and Ethics

A. External Validity

HybridBench is synthetic and controlled. Its opaque answers and exact evidence IDs support mechanism diagnosis, but they do not estimate performance on public QA, long-document memory, conversational personalization, or production traffic. The equal eight-task mixture is a design choice, and each class contains only 15 test scenarios. Model ranks and average grounding effects may change under another task distribution, prompt, context budget, or verifier.

Only one deterministic generation was collected per route–question pair under one experiment seed. Bootstrap intervals resample questions; they do not quantify decoding variance, repeated-run variance, or hardware measurement uncertainty. Secondary intervals were not multiplicity-corrected. The policy evaluation replays a complete frozen query-by-route matrix rather than conducting an independent prospective online deployment.

B. Measurement and Hardware Limits

The energy endpoint covers only selected-board power during generation. It excludes CPU retrieval and memory work, routing, parsing, verification, indexing, model loading, uploads, host idle power, network, and cooling. No idle-power subtraction, whole-system measurement, token-normalized efficiency, or carbon conversion is reported. Four T4 boards were used for clean lanes, route order was route-major, and routes were not randomized or interleaved. Board, session, thermal, and model residency effects may therefore confound cross-lane and cross-model energy comparisons. The Granite-4.0-1B matched endpoints are explicitly cross-board.

C. Model and Interface Limits

Different model families use different tokenizers and chat templates. Reference models ran sequentially alone, whereas online tiny/small routes were evaluated under the resident-pair contract. Input and output token counts differ substantially, so this is not an intrinsic architecture-efficiency leaderboard. The frozen Granite-4.0-1B integration emitted a repeated 80-token sequence and is an observed compatibility failure, not a general assessment of the checkpoint.

The citation guard checks syntax and set membership, not semantic entailment. Exact-ID parsing may reject a value whose intended evidence a human could infer. Conversely, a correctly formatted citation can pass the guard without proving that the natural-language claim is semantically entailed beyond this generated benchmark.

D. Protocol and Robustness Limits

The Stage-06 threshold amendment occurred after validation infeasibility was known but before test-label access. It is fully disclosed and prohibits a positive router claim, yet it remains a deviation from the original implementation path. Five nonanchor routes were pilot-pruned. The test-label seal is procedural because collaborators shared repository access, not a cryptographic guarantee.

The robustness study covers only three deployable anchors and four synthetic corruptions. The selected policy and grounded small-model anchor had zero clean success, creating floor effects. The compromise flag detects a known literal marker and should not be generalized to unseen prompt-injection strategies.

E. Ethics and Environmental Claims

The benchmark is programmatically generated and contains no personal information. Memory records are synthetic, and the deletion task exists to test exclusion of a tombstoned identifier. The work does not estimate carbon emissions, avoided emissions, or environmental impact. Such claims would require a broader energy boundary, location- and time-specific electricity data, and a documented allocation method. The responsible interpretation is limited to measured T4 generation-window board energy within the frozen experiment.

F. Artifact Availability

The source repository contains the notebook builders, execution notebooks, exact analysis code, LaTeX sources, derived tables, and figures. The results package pins the Hugging Face release commit, includes 11 checksum-verified Stage-09 artifacts, and records the frozen specification digest. If the dataset repository remains private during review, artifacts must be supplied through an anonymous or reviewer-accessible archive; the manuscript does not assume public visibility.

IX. Conclusion

E2AM-MemRAG tested a simple but frequently skipped premise: before routing among RAG and memory actions, do those actions produce generator-specific, support-qualified quality separation at a measurable cost? In the frozen benchmark, the answer was mixed. An identical grounded route materially improved Granite-4.1-3B and Qwen3-4B, had an uncertain positive effect for SmolLM3, removed the only successful direct stratum for Qwen3-0.6B, and exposed a deterministic integration failure for Granite-4.0-1B. Every grounded endpoint had a higher observed generation-window GPU-board energy mean, with four matched pairs measured on the same board.

The router consequently had no validation-feasible operating point. Its disclosed fail-closed policy used the same fallback for every test query, achieved zero strict success, and reduced energy by exactly zero; the confirmatory hypothesis failed. This negative result is more useful than an optimistic aggregate. It identifies route viability, interface compliance, task-conditioned grounding value, precise measurement boundaries, and nonzero clean utility as prerequisites for credible energy-aware RAG routing.

Future work should repair model-specific interfaces under a new frozen specification, add public and long-context benchmarks, measure whole-system energy with repeated interleaved runs, and train a router only after its calibration split demonstrates a feasible operating point and query-dependent separation among compatible routes. Until then, retrieval and memory should be treated as generator-dependent interventions whose value must be measured, not assumed.

Appendix A
Complete Route Catalog and Hardware Assignment

Table VII reports every predeclared route and identifies the 17 retained clean-test routes. "Retained" does not mean "best": all matched model anchors were mandatory even when pilot quality was weak. The route ID prefix is a stable experimental identifier, not a rank.

Table VIII anonymizes the physical board UUIDs while preserving comparability structure. Routes sharing an alias used the same T4. Separate robustness conditions were executed on separate boards and are not included in cross-condition energy comparisons.

Appendix B
Scoring and Router Contract

A. Token and Citation Metrics

Let $n_t^{\hat{y}}$ and n_t^{y} denote predicted and target token counts. Answer precision, recall, and F1 are

$$P_{\text{ans}} = \frac{\sum_t \min(n_t^{\hat{y}}, n_t^{y})}{|T_{\hat{y}}|}, \tag{8}$$

$$R_{\text{ans}} = \frac{\sum_t \min(n_t^{\hat{y}}, n_t^{y})}{|T_{y}|}, \tag{9}$$

$$F_1 = \frac{2P_{\text{ans}}R_{\text{ans}}}{P_{\text{ans}} + R_{\text{ans}}}. \tag{10}$$

For required IDs R_q and cited IDs C_{qr},

$$R_{\text{cite}} = \frac{|C_{qr} \cap R_q|}{|R_q|}, \qquad P_{\text{cite}} = \frac{|C_{qr} \cap R_q|}{|C_{qr}|}. \tag{11}$$

Recall is one when no evidence is required. Precision for an empty citation set is one only when no evidence is required. Strict success additionally requires $C_{qr} \subseteq G_{qr}$ for retrieved IDs G_{qr} and $F_q \cap (C_{qr} \cup G_{qr}) = \varnothing$ for forbidden IDs F_q.

B. Router Features and Hyperparameters

The eight query-only features are token count, character count, digit count, entity-code count, temporal-term count, conflict-term count, memory-term count, and multi-hop-term count. The eight probe features are document top score, document score gap, document count, memory top score, memory score gap, memory count, maximum authority, and authority range.

Each bootstrap model uses histogram gradient boosting with learning rate 0.06, 180 iterations, at most 15 leaves, and $L_2 = 1$. Success is calibrated by isotonic regression; energy and latency use 0.90-quantile loss on log outcomes. Bootstrap seeds are 4622, 1701, 31415, 27182, and 65537.

C. Validation Amendment Record

The frozen policy records the following non-negotiable fields:

- amendment ID: stage06-threshold-completion-separation-v1;
- timing: after validation, before test access;
- validation feasible: false;
- selected threshold: $\tau = 1.0$;
- selection mode: fail-closed-tau-1.0-validation-infeasible;
- positive hypothesis claim permitted: false.

Every tested threshold from 0.60 to 1.00 had 8.75% validation success, 100% execution coverage, and 18.75% abstention. The frozen policy file has SHA-256 fda2bda0a715227162ceead0ff032f2228999f6dc9f4bb22 890ec23a4acdab14 in the Stage-06 manifest.

Appendix C
Reproducibility and Artifact Manifest

A. Frozen Counts and Identifiers

The release identifier is e2am-memrag-v3r1; the execution-spec SHA-256 is 1c8f29fd250b87d3546c6ff3 d128dc3fe6600bc798c3c0b0b8c98e0b95c76cfc. The visible Hugging Face release commit is 0b2405d9cca43fd04e35f792 fdc4664405154fc6; the paper-release branch commit is 00fa353f273f3a4b3d57a0b998301c85a1bc098b. Eleven Stage-09 artifacts totaling 11,528,142 bytes passed byte and SHA-256 verification.

The benchmark contains 5,600 documents and 2,400 memory events across 800 scenario groups. After tombstone processing, one selected-seed memory per scenario remains active. The clean release contains 17 routes, 120 query IDs, and 2,040 rows; the robustness release contains three routes, four conditions, 120 query IDs, and 1,440 rows.

B. Rebuild Procedure

The repository script scripts/build_manuscript_assets.py reads only the checksum-verified release and deterministically regenerates manuscript CSVs, tables, and vector figures. scripts/validate_manuscript.py checks numeric claims, bibliography keys, references, measurement-boundary language, artifact existence, and protocol-amendment disclosure. The complete source is compiled with Tectonic 0.16.9 using paper/manuscript/main.tex; the same sources are compatible with Overleaf/pdfLaTeX.

C. Availability

Code and paper sources are hosted at https://github.com/Shanmuk4622/E2AM-MemRAG. Frozen traces reside in the Hugging Face dataset Shanmuk4622/E2AM-MemRAG-Traces. Repository visibility must be checked at submission time. If the trace repository is not public, the authors should provide a checksum-identical anonymous archive or grant reviewer access rather than claiming unrestricted availability.

D. Author Contribution and Conflicts

Shanmukesh Bonala conceived the study, implemented the benchmark and pipeline, executed and audited the experiments, analyzed the data, and wrote the manuscript. The author declares no competing interests and no external funding for this study.

TABLE VII
Predeclared route catalog and pilot retention. A05–A07, A11, and A15 were removed only by the frozen pilot rule; all mandatory matched endpoints were retained.

ID	Generator	Mechanism	Test retained	Success	GPU J
A00	Qwen3 0.6B	Direct	yes	12.5%	54.71
A01	Qwen3 0.6B	BM25 knowledge	yes	0.0%	157.68
A02	Qwen3 0.6B	Dense knowledge	yes	0.0%	150.41
A03	Qwen3 0.6B	Bounded hybrid knowledge	yes	0.0%	143.16
A04	Granite 4.0 1B	Direct	yes	0.0%	256.98
A05	Granite 4.0 1B	BM25 knowledge	no	–	–
A06	Granite 4.0 1B	Dense knowledge	no	–	–
A07	Granite 4.0 1B	Bounded hybrid knowledge	no	–	–
A08	Qwen3 0.6B	Flat memory	yes	0.0%	148.96
A09	Qwen3 0.6B	Hierarchical memory	yes	0.0%	170.32
A10	Qwen3 0.6B	Graph/temporal memory	yes	0.0%	171.49
A11	Granite 4.0 1B	Graph/temporal memory	no	–	–
A12	Granite 4.0 1B	Hybrid knowledge + flat memory	yes	0.0%	258.54
A13	Granite 4.0 1B	Hybrid + graph memory + guard	yes	0.0%	260.85
A14	Qwen3 4B Instruct	Hybrid + graph memory + guard	yes	42.5%	256.13
A15	Granite 4.0 1B	BM25 + hierarchical memory + guard	no	–	–
M16	Qwen3 0.6B	Matched grounded/verified	yes	0.0%	201.44
M17	Granite 4.1 3B	Matched direct	yes	20.8%	57.14
M18	Granite 4.1 3B	Matched grounded/verified	yes	44.2%	176.68
M19	SmolLM3 3B	Matched direct	yes	23.3%	107.42
M20	SmolLM3 3B	Matched grounded/verified	yes	34.2%	148.93
M21	Qwen3 4B Instruct	Matched direct	yes	13.3%	84.42

TABLE VIII
Anonymized physical-T4 assignment for clean-test routes. Within-board contrasts avoid board identity as a direct confound; cross-board energy contrasts remain descriptive.

Route	Physical board alias
A00_tiny_direct	GPU-D
A01_tiny_bm25	GPU-D
A02_tiny_dense	GPU-D
A03_tiny_hybrid	GPU-D
A04_small_direct	GPU-C
A08_tiny_memory_flat	GPU-A
A09_tiny_memory_hier	GPU-A
A10_tiny_memory_graph	GPU-A
A12_small_hybrid_both	GPU-B
A13_small_hybrid_verified	GPU-B
A14_upper_hybrid_verified	GPU-B
M16_tiny_grounded_verified	GPU-D
M17_granite_direct	GPU-D
M18_granite_grounded_verified	GPU-D
M19_peer_direct	GPU-A
M20_peer_grounded_verified	GPU-A
M21_upper_direct	GPU-B

Appendix D
Scenario-Specific Grounding Effects

Table IX expands the aggregate matched effects into the eight 15-query test strata. It shows why an always-grounded average must not be interpreted as the conditional effect of retrieval when retrieval is required.

Acknowledgment

The author thanks the open-source model and tooling communities whose released artifacts made the controlled study possible. No external sponsor influenced the study design, analysis, or reporting.

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TABLE IX
Direct → grounded strict success by task class (%). NR: no retrieval; KO: knowledge only; MO: memory only; KM: knowledge plus memory; TU: temporal update; AC: authority conflict; MH: multi-hop; DM: deleted/missing.

Generator	NR	KO	MO	KM	TU	AC	MH	DM
Qwen3 0.6B	100→0	0→0	0→0	0→0	0→0	0→0	0→0	0→0
Granite 4.0 1B	0→0	0→0	0→0	0→0	0→0	0→0	0→0	0→0
Granite 4.1 3B	66.7→0	0→93.3	0→100	0→20.0	0→100	0→40.0	0→0	100→0
SmolLM3 3B	86.7→0	0→20.0	0→100	0→13.3	0→93.3	0→33.3	0→0	100→13.3
Qwen3 4B	6.7→0	0→93.3	0→66.7	0→13.3	0→100	0→40.0	0→26.7	100→0

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